DIFFERENTIAL EQUATIONS

1.	The sum of the order and the degree of the differential equation $\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^3 = \sin y$ is : (a) 5 (b) 2 (c) 3 (d) 4
2.	The general solution of the differential equation $x\,dy - (1 + x^2)\,dx = dx$ is : (a) $y = 2x + \frac{x^3}{3} + C$ (b) $y = 2\log x + \frac{x^3}{3} + C$ (c) $y = \frac{x^2}{2} + C$ (d) $y = 2\log x + \frac{x^2}{2} + C$
3.	Find the particular solution of the differential equation $\frac{dy}{dx} + \sec^2 x \,.\, y = \tan x \,.\, \sec^2 x$, given that $y(0) = 0$.
4.	**Solve the differential equation given by $x\,dy - y\,dx - \sqrt{x^2 + y^2}\;dx = 0$.**
5.	Degree of the differential equation $\sin x + \cos\left(\frac{dy}{dx}\right) = y^2$ is (A) 2 (B) 1 (C) not defined (D) 0
6.	The integrating factor of the differential equation $(1 - y^2)\frac{dx}{dy} + yx = ay, \ (-1 < y < 1)$ is (A) $\frac{1}{y^2 - 1}$ (B) $\frac{1}{\sqrt{y^2 - 1}}$ (C) $\frac{1}{1 - y^2}$ (D) $\frac{1}{\sqrt{1 - y^2}}$
7.	Anti-derivative of $\frac{\tan x - 1}{\tan x + 1}$ with respect to x is : (A) $\sec^2\left(\frac{\pi}{4} - x\right) + c$ (B) $-\sec^2\left(\frac{\pi}{4} - x\right) + c$ (C) $\log\left\|\sec\left(\frac{\pi}{4} - x\right)\right\| + c$ (D) $-\log\left\|\sec\left(\frac{\pi}{4} - x\right)\right\| + c$

8.	Find the general solution of the differential equation : $(xy - x^2)\, dy = y^2\, dx$.
9.	Find the general solution of the differential equation : $$(x^2 + 1)\frac{dy}{dx} + 2xy = \sqrt{x^2 + 4}$$
10.	What is the product of the order and degree of the differential equation $\frac{d^2y}{dx^2}\sin y + \left(\frac{dy}{dx}\right)^3 \cos y = \sqrt{y}$? (a) 3 (b) 2 (c) 6 (d) not defined
11.	Find the general solution of the differential equation : $$\frac{d}{dx}(xy^2) = 2y(1 + x^2)$$
12.	Solve the following differential equation : $$x e^{\frac{y}{x}} - y + x\frac{dy}{dx} = 0$$
13.	The integrating factor for solving the differential equation $x\frac{dy}{dx} - y = 2x^2$ is : (a) e^{-y} (b) e^{-x} (c) x (d) $\frac{1}{x}$
14.	The order and degree (if defined) of the differential equation, $\left(\frac{d^2y}{dx^2}\right)^2 + \left(\frac{dy}{dx}\right)^3 = x \sin\left(\frac{dy}{dx}\right)$ respectively are : (a) 2, 2 (b) 1, 3 (c) 2, 3 (d) 2, degree not defined
15.	Find the particular solution of the differential equation $$\frac{dy}{dx} = \frac{x + y}{x}, \quad y(1) = 0.$$

16.	Find the general solution of the differential equation $e^x \tan y \, dx + (1 - e^x) \sec^2 y \, dy = 0$.
17.	The sum of the order and the degree of the differential equation $\frac{d}{dx}\left(\left(\frac{dy}{dx}\right)^3\right)$ is (a) 2 (b) 3 (c) 5 (d) 0
18.	The number of arbitrary constants in the general solution of the differential equation $\frac{dy}{dx} + y = 0$ is : (A) 0 (B) 1 (C) 2 (D) 3
19.	Find the general solution of the differential equation $y \, dx - x \, dy + (x \log x) \, dx = 0$.
20.	The difference of the order and the degree of the differential equation $\left(\frac{d^2y}{dx^2}\right)^2 + \left(\frac{dy}{dx}\right)^3 + x^4 = 0$ is : (a) 1 (b) 2 (c) -1 (d) 0
21.	The integrating factor of the differential equation $(3x^2 + y)\frac{dx}{dy} = x$ is (a) $\frac{1}{x}$ (b) $\frac{1}{x^2}$ (c) $\frac{2}{x}$ (d) $-\frac{1}{x}$
22.	Find the particular solution of the differential equation $\frac{dy}{dx} = \frac{xy}{x^2 + y^2}$, given that $y = 1$ when $x = 0$.
23.	Find the particular solution of the differential equation $(1 + x^2)\frac{dy}{dx} + 2xy = \frac{1}{1 + x^2}$, given that $y = 0$ when $x = 1$.

24.	The integrating factor of the differential equation $(1 - x^2)\frac{dy}{dx} + xy = ax$, $-1 < x < 1$, is : (A) $\frac{1}{x^2 - 1}$ (B) $\frac{1}{\sqrt{x^2 - 1}}$ (C) $\frac{1}{1 - x^2}$ (D) $\frac{1}{\sqrt{1 - x^2}}$
25.	The order and degree of the differential equation $\left[1 + \left(\frac{dy}{dx}\right)^2\right]^3 = \frac{d^2y}{dx^2}$ respectively are : (A) 1, 2 (B) 2, 3 (C) 2, 1 (D) 2, 6
26.	Find the particular solution of the differential equation given by $x^2 \frac{dy}{dx} - xy = x^2 \cos^2\left(\frac{y}{2x}\right)$, given that when $x = 1$, $y = \frac{\pi}{2}$.
27.	The number of solutions of differential equation $\frac{dy}{dx} - y = 1$, given that $y(0) = 1$, is : (A) 0 (B) 1 (C) 2 (D) infinitely many
28.	The degree of the differential equation $(y'')^2 + (y')^3 = x \sin(y')$ is : (A) 1 (B) 2 (C) 3 (D) not defined
29.	Find the particular solution of the differential equation given by $2xy + y^2 - 2x^2 \frac{dy}{dx} = 0$; $y = 2$, when $x = 1$.
30.	Find the general solution of the differential equation : $y\,dx = (x + 2y^2)\,dy$

31.	$x \log x \dfrac{dy}{dx} + y = 2 \log x$ is an example of a : (A) variable separable differential equation. (B) homogeneous differential equation. (C) first order linear differential equation. (D) differential equation whose degree is not defined.
32.	The general solution of the differential equation $x\,dy + y\,dx = 0$ is : (A) $xy = c$ (B) $x + y = c$ (C) $x^2 + y^2 = c^2$ (D) $\log y = \log x + c$
33.	The integrating factor of the differential equation $(x + 2y^2)\dfrac{dy}{dx} = y\ (y > 0)$ is : (A) $\dfrac{1}{x}$ (B) x (C) y (D) $\dfrac{1}{y}$
34.	Find the particular solution of the differential equation $\dfrac{dy}{dx} = y \cot 2x$, given that $y\left(\dfrac{\pi}{4}\right) = 2$.
35.	Find the particular solution of the differential equation $(xe^{\frac{y}{x}} + y)\,dx = x\,dy$, given that $y = 1$ when $x = 1$.
36.	The order of the differential equation $\dfrac{d^4y}{dx^4} - \sin\left(\dfrac{d^2y}{dx^2}\right) = 5$ is : (A) 4 (B) 3 (C) 2 (D) not defined

37. Find the particular solution of the differential equation

$$\frac{dy}{dx} - 2xy = 3x^2 e^{x^2} ; y(0) = 5.$$

38. Solve the following differential equation :

$$x^2 \, dy + y(x + y) \, dx = 0$$

39. Which of the following is <u>not</u> a homogeneous function of x and y ?

(A) $y^2 - xy$ (B) $x - 3y$

(C) $\sin^2 \frac{y}{x} + \frac{y}{x}$ (D) $\tan x - \sec y$

40. The integrating factor of differential equation $(x + 2y^3) \frac{dy}{dx} = 2y$ is

(A) $e^{\frac{y^2}{2}}$ (B) $\frac{1}{\sqrt{y}}$

(C) $\frac{1}{y^2}$ (D) $e^{-\frac{1}{y^2}}$

41. If p and q are respectively the order and degree of the differential equation $\frac{d}{dx}\left(\frac{dy}{dx}\right)^3 = 0$, then (p – q) is

(A) 0 (B) 1

(C) 2 (D) 3

42. Solve the differential equation $2(y + 3) - xy \frac{dy}{dx} = 0$; given $y(1) = -2$.

43. Solve the following differential equation :

$$(1 + x^2)\frac{dy}{dx} + 2xy = 4x^2.$$

44.	The solution of the differential equation $\frac{dy}{dx} = \frac{-x}{y}$ represents family of : (A) parabolas (B) circles (C) ellipses (D) hyperbolas
45.	The integrating factor of the differential equation $\left(\frac{e^{-2\sqrt{x}}}{\sqrt{x}} - \frac{y}{\sqrt{x}}\right)\frac{dx}{dy} = 1$ is : (A) $e^{-1/\sqrt{x}}$ (B) $e^{2/\sqrt{x}}$ (C) $e^{2\sqrt{x}}$ (D) $e^{-2\sqrt{x}}$
46.	The sum of the order and degree of the differential equation $\left[1+\left(\frac{dy}{dx}\right)^2\right]^3 = \frac{d^2y}{dx^2}$ is : (A) 2 (B) $\frac{5}{2}$ (C) 3 (D) 4
47.	Find the particular solution of the differential equation $\left[x \sin^2\left(\frac{y}{x}\right) - y\right] dx + x\, dy = 0$ given that $y = \frac{\pi}{4}$, when $x = 1$.
48.	The order and degree of the differential equation $\frac{d^2y}{dx^2} + 4\left(\frac{dy}{dx}\right) = x \log\left(\frac{d^2y}{dx^2}\right)$ are respectively : (A) 0, 3 (B) 2, 1 (C) 2, not defined (D) 1, not defined
49.	The solution for the differential equation $\log\left(\frac{dy}{dx}\right) = 3x + 4y$ is : (A) $3e^{4y} + 4e^{-3x} + C = 0$ (B) $e^{3x+4y} + C = 0$ (C) $3e^{-3y} + 4e^{4x} + 12C = 0$ (D) $3e^{-4y} + 4e^{3x} + 12C = 0$

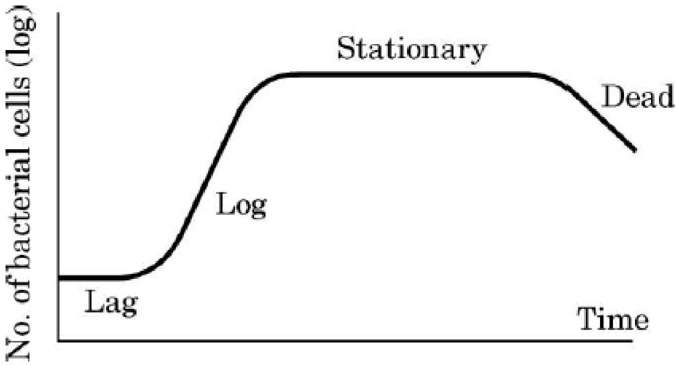
50. Solve the differential equation : $x^2y\,dx - (x^3 + y^3)\,dy = 0$.

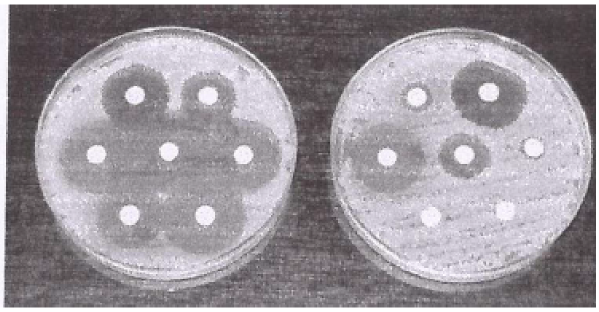
51. Solve the differential equation $(1 + x^2)\dfrac{dy}{dx} + 2xy - 4x^2 = 0$ subject to initial condition $y(0) = 0$.

52. Solve the differential equation $(x - \sin y)\,dy + (\tan y)\,dx = 0$, given $y(0) = 0$.

CASE STUDY

53. A bacteria sample of certain number of bacteria is observed to grow exponentially in a given amount of time. Using exponential growth model, the rate of growth of this sample of bacteria is calculated.





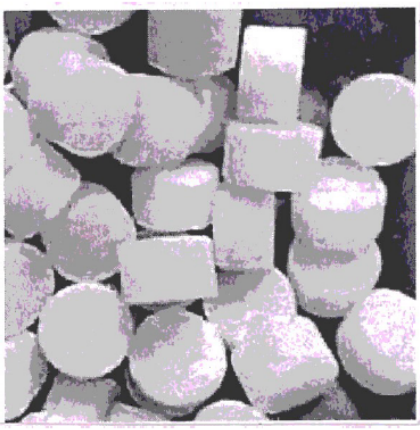
The differential equation representing the growth of bacteria is given as :

$\dfrac{dP}{dt} = kP$, where P is the population of bacteria at any time 't'.

Based on the above information, answer the following questions :

(i) Obtain the general solution of the given differential equation and express it as an exponential function of 't'.

(ii) If population of bacteria is 1000 at $t = 0$, and 2000 at $t = 1$, find the value of k.

54.	An equation involving derivatives of the dependent variable with respect to the independent variables is called a differential equation. A differential equation of the form $\frac{dy}{dx} = F(x, y)$ is said to be homogeneous if F(x, y) is a homogeneous function of degree zero, whereas a function F(x, y) is a homogenous function of degree n if $F(\lambda x, \lambda y) = \lambda^n F(x, y)$. To solve a homogeneous differential equation of the type $\frac{dy}{dx} = F(x, y) = g\left(\frac{y}{x}\right)$, we make the substitution $y = vx$ and then separate the variables. Based on the above, answer the following questions : (I) Show that $(x^2 - y^2)\, dx + 2xy\, dy = 0$ is a differential equation of the type $\frac{dy}{dx} = g\left(\frac{y}{x}\right)$. (II) Solve the above equation to find its general solution.
55.	Camphor is a waxy, colourless solid with strong aroma that evaporates through the process of sublimation, if left in the open at room temperature.  (Cylindrical-shaped Camphor tablets) A cylindrical camphor tablet whose height is equal to its radius (r) evaporates when exposed to air such that the rate of reduction of its volume is proportional to its total surface area. Thus, $\frac{dV}{dt} = kS$ is the differential equation, where V is the volume, S is the surface area and t is the time in hours.

Based upon the above information, answer the following questions :

(i) Write the order and degree of the given differential equation. *1*

(ii) Substituting $V = \pi r^3$ and $S = 2\pi r^2$, we get the differential equation $\frac{dr}{dt} = \frac{2}{3}k$. Solve it, given that $r(0) = 5$ mm. *1*

(iii) (a) If it is given that $r = 3$ mm when $t = 1$ hour, find the value of k. Hence, find t for $r = 0$ mm. *2*

OR

(iii) (b) If it is given that $r = 1$ mm when $t = 1$ hour, find the value of k. Hence, find t for $r = 0$ mm. *2*

56. During a heavy gaming session, the temperature of a student's laptop processor increases significantly. After the session, the processor begins to cool down, and the rate of cooling is proportional to the difference between the processor's temperature and the room temperature (25°C). Initially the processor's temperature is 85°C. The rate of cooling is defined by the equation $\frac{d}{dt}(T(t)) = -k(T(t) - 25)$,

where $T(t)$ represents the temperature of the processor at time t (in minutes) and k is a constant.

Based on the above information, answer the following questions :

(i) Find the expression for temperature of processor, $T(t)$ given that $T(0) = 85°C$. *2*

(ii) How long will it take for the processor's temperature to reach 40°C ? Given that $k = 0{\cdot}03$, $\log_e 4 = 1{\cdot}3863$. *2*